FOUNTATIONS OF AI

WEEK 1 & 2 NOTES

Definition of Artificial Intelligence (AI)

Artificial Intelligence (AI is a branch of Computer Science that focuses on the design, development, and deployment of intelligent systems capable of performing tasks that normally require human intelligence. These tasks include learning from experience, reasoning and problem-solving, understanding natural language, recognizing patterns in images or speech, making decisions, and adapting to new situations.

AI systems are designed to simulate cognitive functions such as perception, memory, and logical thinking. Unlike traditional computer programs that follow strictly predefined rules, AI-powered systems can analyze large volumes of data, identify patterns, and improve their performance over time through techniques such as machine learning and deep learning.

AI encompasses several subfields, including machine learning, natural language processing, computer vision, robotics, and expert systems. These technologies enable applications such as virtual assistants, recommendation systems, autonomous vehicles, fraud detection systems, medical diagnosis tools, and climate forecasting models.

Examples of AI Applications

Healthcare: Predicting diseases and personalizing treatments.

Finance: Fraud detection and algorithmic trading.

Retail: Recommendation systems for personalized shopping.

Technologies: TensorFlow, PyTorch, and Scikit-learn will remain central tools for ML development.

APPROACHES TO ARTIFICIAL INTELLIGENCE

Artificial Intelligence is commonly categorized into four main approaches depending on two key dimensions. Whether it aims to follow ideal logical reasoning or achieve the best possible outcome. The approaches to AI are categorized as follows;

1. **Thinking Humanly (Cognitive Modeling Approach) -** *Can machines think the way humans think?*

This approach focuses on building systems that **replicate how humans actually think**.

- It is grounded in psychology and cognitive science.
- Researchers study how humans solve problems, make decisions, and process information.
- The goal is to model those internal mental processes in machines.

For example, artificial neural networks are inspired by the structure of the human brain. Instead of just producing correct answers, the system attempts to simulate the same thinking patterns human's use.

2. Thinking Rationally (The "Laws of Thought" Approach) - *Can machines reason correctly using formal logic?*

This approach is based on formal logic.

It emphasizes correct reasoning according to logical rules.

Systems are designed to follow structured reasoning processes to arrive at valid conclusions.

It relies heavily on mathematics, symbolic logic, and rule-based systems.

Rather than copying how humans think (which can sometimes be biased or irrational), this approach focuses on ideal reasoning what humans *should* think if they were perfectly logical.

3. Acting Humanly (The Turing Test Approach) - *Can a machine act so convincingly human that we cannot tell the difference?*

This approach evaluates AI based on whether it can behave like a human.

It is associated with Alan Turing, who proposed the Turing Test.

In this test:

- A human interacts with both a machine and another human through text.
- If the evaluator cannot reliably distinguish the machine from the human, the machine is considered intelligent.
- To pass the Turing Test, a system needs:
- Natural Language Processing (to communicate),
- Knowledge Representation,

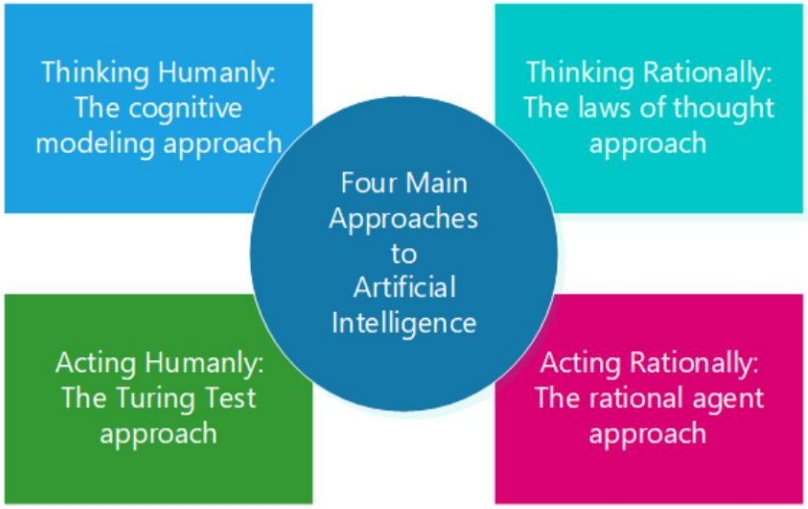
4. Acting Rationally (The Rational Agent Approach)

This is the most widely accepted modern approach to AI.

Instead of focusing on human imitation, it focuses on **optimal performance**.

A rational agent:

- Perceives its environment,
- Makes decisions based on available information,
- Acts in a way that maximizes its chances of achieving its goals.



AI approaches summary as figure above (FOUR MAIN APPROACHES OF AI)

1. *Acting Humanly: The Turing Test approach (Acting like a human).*
2. *Thinking Humanly: The cognitive modeling approach (Thinking like a person).*
3. *Thinking Rationally: The laws of thought approach (Thinking rationally is a logica)l process and it concludes based on symbolic logic. (The famous syllogisms, "Socrates is a man; all men are mortal; therefore, Socrates is mortal.")*
4. *Acting Rationally: The rational agent approach (Rational agent acts to achieve high value and brings the best possible outcome for any given task).*

UNDERSTANDING AI AGENTS

An artificial intelligence (AI) agent is a software program that can interact with its environment, collect data, and use that data to perform self-directed tasks that meet predetermined goals. Humans set goals, but an AI agent independently chooses the best actions it needs to perform to achieve those goals. For example, consider a contact center AI agent that wants to resolve customer queries. The agent will automatically ask the customer different questions, look up information in internal documents, and respond with a solution. Based on the customer responses, it determines if it can resolve the query itself or pass it on to a human.

Multiple AI agents can collaborate to automate complex workflows and can also be used in agentic ai systems. They exchange data with each other, allowing the entire system to work together to achieve common goals. Individual AI agents can be specialized to perform specific subtasks with accuracy. An orchestrator agent coordinates the activities of different specialist agents to complete larger, more complex tasks.

What are the key principles that define AI agents?

All software autonomously performs various routine tasks as specified by the software developer. So, what makes AI agents special?

Autonomy

AI agents act autonomously, without constant human intervention. While traditional software follows hard-coded instructions, AI agents identify the next appropriate action based on past data and execute it without continuous human oversight.

For example, a bookkeeping agent automatically flags and requests missing invoice data for purchases.

Goal-oriented behavior

AI agents are driven by objectives. Their actions aim to maximize success as defined by a utility function or performance metric. Unlike traditional programs that merely complete tasks, intelligent agents pursue goals and evaluate the consequences of their actions in relation to those goals.

For example, an AI logistics system optimizes delivery routes to balance speed, cost, and fuel consumption simultaneously, thereby balancing multiple objectives.

Perception

AI agents interact with their environment by collecting data through sensors or digital inputs. They can collect data from external systems and tools via APIS. This data allows them to perceive the world around them, recognize changes, and update their internal state accordingly.

For example, cybersecurity agents collect data from third-party databases to remain aware of the latest security incidents.

Rationality

AI agents are rational entities with reasoning capabilities. They combine data from their environment with domain knowledge and past context to make informed decisions, achieving optimal performance and results.

For example, a robotic agent collects sensor data, and a chatbot uses customer queries as input. The AI agent applies the data to make an informed decision. It analyzes the collected data to predict the best outcomes that support predetermined goals. The agent also uses the results to formulate the next action that it should take. For example, self-driving cars navigate around obstacles on the road based on data from multiple sensors.

Proactivity

AI agents can take initiative based on forecasts and models of future states. Instead of simply reacting to inputs, they anticipate events and prepare accordingly.

For instance, an AI-based customer service agent might reach out to a user whose behavior suggests frustration, offering help before a support ticket is filed. Autonomous warehouse robots may reposition themselves in anticipation of upcoming high-traffic operations.

Continuous learning

AI agents improve over time by learning from past interactions. They identify patterns, feedback, and outcomes to refine their behavior and decision-making. This differentiates them from static programs that always behave the same way regardless of new inputs.

For instance, predictive maintenance agents learn from past equipment failures to better forecast future issues.

Adaptability

AI agents adjust their strategies in response to new circumstances. This flexibility allows them to handle uncertainty, novel situations, and incomplete information.

For example, a stock trading bot adapts its strategy during a market crash, while a game-playing agent like AlphaZero discovers new tactics through self-play, even without prior human strategies.

Collaboration

AI agents can work with other agents or human agents to achieve shared goals. They are capable of communicating, coordinating, and cooperating to perform tasks together. Their collaborative

behavior often involves negotiation, sharing information, allocating tasks, and adapting to others' actions.

For example, multi-agent systems in healthcare can have agents specializing in specific tasks like diagnosis, preventive care, medicine scheduling, etc., for holistic patient care automation.

What are the benefits of using AI agents?

AI agents can improve your business operations and your customers' experiences.

Improved productivity

Business teams are more productive when they delegate repetitive tasks to AI agents. This way, they can divert their attention to mission-critical or creative activities, adding more value to their organization.

Reduced costs

Businesses can utilize intelligent agents to minimize unnecessary costs resulting from process inefficiencies, human errors, and manual processes. They can confidently tackle complex tasks because autonomous agents follow a consistent model that adapts to changing environments. Agent technology automating business processes can lead to significant cost savings.

Informed decision-making

Advanced intelligent agents have predictive capabilities and can collect and process massive amounts of real-time data. This enables business managers to make more informed predictions at speed when strategizing their next move. For example, you can use AI agents to analyze product demands in different market segments when running an ad campaign.

Improved customer experience

Customers seek engaging and personalized experiences when interacting with businesses. Integrating AI agents allows businesses to personalize product recommendations, provide prompt responses, and innovate to improve customer engagement, conversion, and loyalty. AI agents can provide detailed responses to complex customer questions and resolve challenges more efficiently.

What are the key components of AI agent architecture?

An AI agent architecture contains the following key components.

Foundation model

At the core of any AI agent lies a foundation or large language model (LLM) such as GPT or Claude. It enables the agent to interpret natural language inputs, generate human-like responses, and reason over complex instructions. The LLM acts as the agent's reasoning engine, processing prompts and transforming them into actions, decisions, or queries to other components (e.g.,

memory or tools). It retains some memory across sessions by default and can be coupled with external systems to simulate continuity and context awareness.

Planning module

The planning module enables the agent to break down goals into smaller, manageable steps and sequence them logically. This module employs symbolic reasoning, decision trees, or algorithmic strategies to determine the most effective approach for achieving a desired outcome. It can be implemented as a prompt-driven task decomposition or more formalized approaches, such as Hierarchical Task Networks (HTNs) or classical planning algorithms. Planning allows the agent to operate over longer time horizons, considering dependencies and contingencies between tasks.

Memory module

The memory module allows the agent to retain information across interactions, sessions, or tasks. This includes both short-term memories, such as chat history or recent sensor input, and long-term memory, including customer data, prior actions, or accumulated knowledge. Memory enhances the agent's personalization, coherence, and context-awareness. When building AI agents, developers use vector databases or knowledge graphs to store and retrieve semantically meaningful content.

Tool integration

AI agents often extend their capabilities by connecting to external software, APIs, or devices. This allows them to act beyond natural language, performing real-world tasks such as retrieving data, sending emails, running code, querying databases, or controlling hardware. The agent identifies when a task requires a tool and then delegates the operation accordingly. Tool use is typically guided by the LLM through planning and parsing modules that format the tool call and interpret its output.

Learning and reflection

Reflection can occur in multiple forms:

- The agent evaluates the quality of its own output (e.g., did it solve the problem correctly?).
- Human users or automated systems provide corrections.
- The agent selects uncertain or informative examples to improve its learning.

Reinforcement Learning (RL) is a key learning paradigm. The agent interacts with an environment, receives feedback in the form of rewards or penalties, and learns a policy that maps states to actions for maximum cumulative reward. RL is especially useful in environments where explicit training data is sparse, such as robotics, gaming, or financial trading. The agent balances

exploration (trying new actions) and exploitation (using known best actions) to improve its strategy over time.

How does an AI agent work?

AI agents work by simplifying and automating complex tasks. Most autonomous agents follow a specific workflow when performing assigned tasks.

Determine goals

The AI agent receives a specific instruction or goal from the user. It uses the goal to plan tasks that make the final outcome relevant and useful to the user. Then, the agent breaks down the goal into several smaller, actionable tasks. To achieve the goal, the agent performs those tasks based on specific orders or conditions.

Acquire information

AI agents require information to execute tasks they have planned successfully. For example, the agent must extract conversation logs to analyze customer sentiments. As such, AI agents might access the internet to search for and retrieve the information they need. In some applications, an intelligent agent can interact with other agents or machine learning models to access or exchange information.

Implement tasks

With sufficient data, the AI agent methodically implements the task at hand. Once it accomplishes a task, the agent removes it from the list and proceeds to the next one. Between task completions, the agent evaluates whether it has achieved the designated goal by seeking external feedback and inspecting its own logs. During this process, the agent may create and act on additional tasks to achieve the final outcome.

What are the types of AI agents?

Organizations create and deploy AI agents across a range of types and tasks. We share some examples below.

Simple reflex agents

A simple reflex agent operates strictly based on predefined rules and its immediate data. It will not respond to situations beyond a given event, condition, and action rule. Hence, these agents are suitable for simple tasks that don’t require extensive training. For example, you can use a simple reflex agent to reset passwords by detecting specific keywords in a user’s conversation.

Model-based reflex agents

A model-based agent is similar to simple reflex agents, except that it has a more advanced decision-making mechanism. Rather than merely following a specific rule, a model-based agent

evaluates probable outcomes and consequences before making a decision. Using supporting data, it builds an internal model of the world it perceives and uses that to support its decisions.

Goal-based agents

Goal-based agents, also known as rule-based agents, are AI agents that possess more robust reasoning capabilities. Besides evaluating the environment data, the agent compares different approaches to help it achieve the desired outcome. Goal-based agents always choose the most efficient path. They are suitable for performing complex tasks, such as natural language processing (NLP) and robotics applications.

Utility-based agents

A utility-based agent employs a complex reasoning algorithm to assist users in maximizing the outcome they desire. The agent compares different scenarios and their respective utility values or benefits. Then, it selects one that offers users the most rewards. For example, customers can use a utility-based agent to search for flight tickets with the minimum travel time, regardless of the price.

Learning agents

A learning agent continually learns from past experiences to enhance its performance. Using sensory input and feedback mechanisms, the agent adapts its learning element over time to meet specific standards. Additionally, it utilizes a problem generator to design new tasks that train itself using collected data and past results.

Hierarchical agents

Hierarchical agents are an organized group of intelligent agents arranged in tiers. Higher-level agents decompose complex tasks into smaller ones and assign them to lower-level agents. Each agent runs independently and submits a progress report to its supervising agent. The higher-level agent collects the results and coordinates subordinate agents to ensure they collectively achieve goals.

Multi-agent systems

A multi-agent system (MAS) consists of multiple agents that interact with one another to solve problems or achieve shared objectives. These agents can be homogeneous (similar in design) or heterogeneous (different in structure or function) and may collaborate, coordinate, or even compete depending on the context. MAS are particularly effective in complex, distributed environments where centralized control is impractical.

For example, in autonomous vehicle fleets, each vehicle acts as an independent agent but collaborates with others to avoid traffic congestion and prevent collisions, leading to smoother traffic flow.

What are the challenges of using AI agents?

AI agents are helpful software technologies that automate business workflows to achieve better outcomes. That being said, organizations should address the following concerns when deploying autonomous AI agents for business use cases.

Data privacy concerns

Developing and operating advanced AI agents requires acquiring, storing, and moving massive volumes of data. Organizations should be aware of data privacy requirements and employ necessary measures to improve their data security posture.

Ethical challenges

In certain circumstances, AI models may produce results that are biased or inaccurate. Applying safeguards, such as human reviews, helps to ensure customers receive helpful and fair responses from the agents deployed.

Technical complexities

Implementing advanced AI agents requires specialized experience and knowledge of machine learning technologies. Developers must be able to integrate machine learning libraries with software applications and train the agent with enterprise-specific data.

Limited compute resources

Training and deploying deep learning AI agents require substantial computing resources. When organizations implement these agents on-premise, they must invest in and maintain costly infrastructure that is not easily scalable.

Agents and Multi Agent Systems for AI Solutions

What is an AI AGENT

AI agents are software systems that use AI to pursue goals and complete tasks on behalf of users. They show reasoning, planning, and memory and have a level of autonomy to make decisions, learn, and adapt.

Their capabilities are made possible in large part by the multimodal capacity of generative AI and AI foundation models. AI agents can process multimodal information like text, voice, video, audio, code, and more simultaneously; can converse, reason, learn, and make decisions. They can learn over time and facilitate transactions and business processes. Agents can work with other agents to coordinate and perform more complex workflows.

Multi-Agent System in AI

A multi-agent system in AI (or MAS for short) is just as it sounds—separate agents working together to complete tasks. Single agent systems rely on one solitary program to do all the heavy lifting, but a MAS is all about the dynamic interplay between agents.

Each agent is like a mini decision-maker, equipped with its own goals, abilities, and even a bit of personality.

When you rely on a single AI agent to handle every problem, you are almost guaranteeing inefficiency and frustration.

As tasks become more complex and varied, that one agent is forced to manage too many responsibilities at once. Instead of operating effectively, it becomes overloaded — struggling to process competing priorities, adapt to different contexts, and respond to multiple demands simultaneously.

In a modern business environment, where operations are dynamic and interconnected, expecting one AI system to do everything reduces performance and scalability. A more effective approach is to distribute responsibilities across specialized agents, each designed to handle specific functions efficiently and collaboratively.

Generally, AI agents are most powerful when they operate within a shared environment — collaborating when goals align and competing when interests differ much like a sports team or a dynamic city filled with ambitious individuals.

Take a volleyball team as an analogy. The objective of a volleyball game is simple: score points by preventing the ball from hitting the ground on your side of the court and forcing it to land on your opponent's side.

Now imagine the rules stay the same, but your team has only one player. Technically, it's still possible to win but the challenge increases dramatically. That single player must receive, set, and spike every ball alone. The workload is overwhelming, reactions are slower, and strategy is limited.

Contrast that with a full team, where each player specializes in a distinct role: one excels at digging, another at setting, another at powerful spikes. Suddenly, the game becomes coordinated, strategic, and far more effective. Each player focuses on what they do best, and the team as a whole performs at a higher level. That's the essence of a multi-agent system.

COMPONENTS OF AGENT SYSTEMS: AGENT ENVIRONMENTS; AGENTS DESCRIPTION

At its core, a Multi-Agent System (MAS) is built on two fundamental elements:

1. The Agents

Agents are autonomous entities capable of making decisions, learning from experience, and adapting to changes. They interact not only with their environment but also with one another.

Each agent:

- Has its own goals or objectives
- Perceives its surroundings through inputs
- Takes actions based on internal decision-making processes
- May collaborate or compete with other agents

You can think of agents as team members — each bringing unique skills, knowledge, and perspectives to achieve individual or shared goals.

2. The Environment

The environment is the shared space in which agents operate and interact. It provides the context, rules, and conditions that shape their behavior.

The environment:

- Supplies information to agents (through observations or signals)
- Responds to agents' actions
- May change dynamically over time

This environment could be:

- A simulated virtual world
- A computer network
- A marketplace
- A physical setting (such as robots operating in a warehouse)
- Or even a data-driven system like financial markets or weather models

Bringing It Together

The intelligence of a Multi-Agent System emerges from the continuous interaction between agents and their environment. Agents perceive, decide, and act — the environment updates — and the cycle continues.

It is this ongoing feedback loop, combined with agent collaboration and competition, that enables MAS to handle complex, distributed, and dynamic problems more effectively than isolated systems.

Types of agents in AI

AI agents can be categorized in various ways based on their capabilities, roles, and environments. Here are some key categories of agents:

There are different definitions of agent types and agent categories.

Based on interaction

One way to categorize agents is by how they interact with users. Some agents engage in direct conversation, while others operate in the background, performing tasks without direct user input:

1. **Interactive partners** (also known as, surface agents) – Assisting us with tasks like customer service, healthcare, education, and scientific discovery, providing personalized and intelligent support. Conversational agents include Q&A, chit chat, and world knowledge interactions with humans. They are generally user query triggered and fulfill user queries or transactions.
2. **Autonomous background processes** (also known as, background agents) – Working behind the scenes to automate routine tasks, analyze data for insights, optimize processes for efficiency, and proactively identify and address potential issues. They include workflow agents. They have limited or no human interaction and are generally driven by events and fulfill queued tasks or chains of tasks.

Based on number of agents

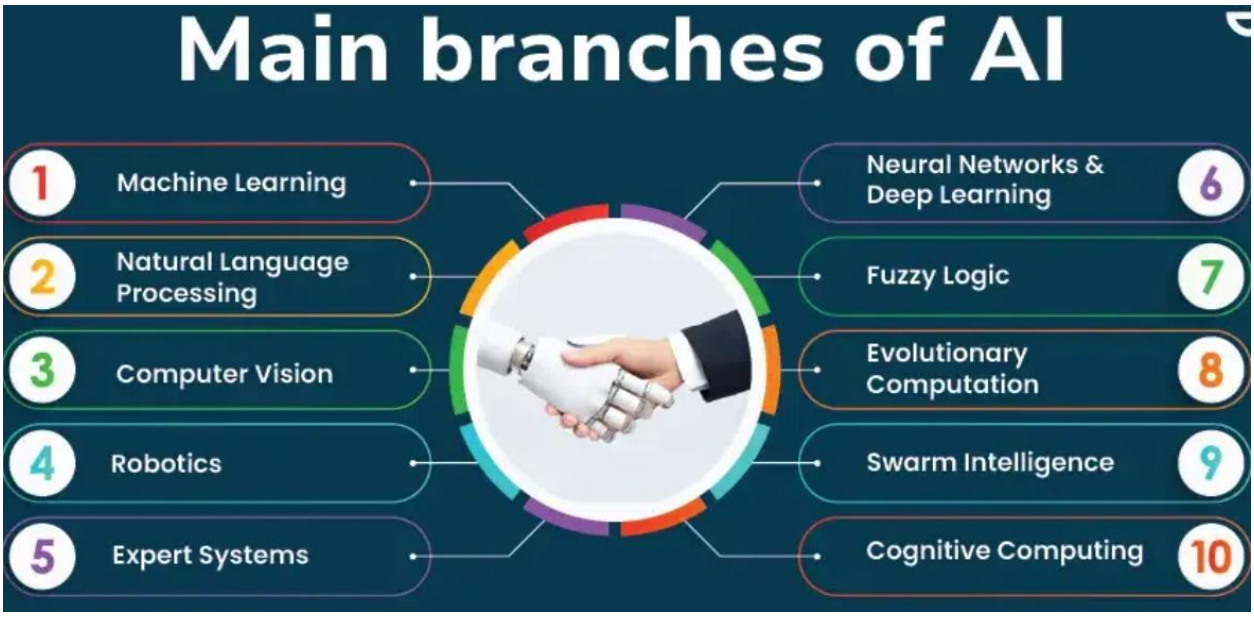
1. **Single agent**: Operate independently to achieve a specific goal. They utilize external tools and resources to accomplish tasks, enhancing their functional capabilities in diverse environments. They are best suited for well defined tasks that do not require collaboration with other AI agents. Can only handle one foundation model for its processing.
2. **Multi-agent**: Multiple AI agents that collaborate or compete to achieve a common objective or individual goals. These systems leverage the diverse capabilities and roles of individual agents to tackle complex tasks. Multi-agent systems can simulate human

behaviors, such as interpersonal communication, in interactive scenarios. Each agent can have different foundation models that best fit their needs.

BRANCHES/FIELDS OF AI

Artificial intelligence (AI*)* is the leading component of innovation and serves as a tool that imitates human thinking. The branches of AI encompass machine learning and auto-robots, which include self-driving cars, smart homes, virtual personal assistants, and other automated systems. These AI systems demonstrate incredible prowess in recognizing and analyzing large sets of data, making decisions, and solving problems.

- ***Machine Learning,***
- ***Natural Language Processing,***
- ***Computer Vision, Robotics,***
- ***Expert Systems,***
- ***Neural Networks and Deep Learning,***
- ***Fuzzy Logic,***
- ***Evolutionary Computation,***
- ***Swarm Intelligence, and Cognitive Computing.***



Main Branches of AI

Artificial Intelligence (AI) is a broad field that encompasses various subfields and branches, each focusing on different aspects of creating intelligent machines. Here are the main branches of AI:

1. <u>Machine Learning (ML)</u>

Machine Learning is the study of computer algorithms that improve automatically through experience and data.

- **Applications**: Image and speech recognition, predictive analytics, recommendation systems.
- **Subfields**: Supervised Learning, Unsupervised Learning, Reinforcement Learning.
- **Key Algorithms**: Decision Trees, Support Vector Machines, Neural Networks, k-Nearest Neighbors.

2. <u>Natural Language Processing (NLP)</u>

- **Definition**: NLP focuses on the interaction between computers and human languages, enabling machines to understand, interpret, and generate human language.
- **Applications**: Language translation, sentiment analysis, chatbots, text summarization.
- **Challenges**: Ambiguity, context understanding, linguistic diversity.
- **Technologies**: Tokenization, Part-of-Speech Tagging, Named Entity Recognition, Machine Translation.

3. <u>Computer Vision</u>

Computer Vision involves enabling machines to interpret and make decisions based on visual data.

- **Applications**: Facial recognition, autonomous vehicles, medical image analysis, surveillance.
- **Technologies**: Image processing, object detection, pattern recognition.
- **Key Techniques**: Convolutional Neural Networks (CNNs), Image Segmentation, Feature Extraction.

4. <u>Robotics</u>

Robotics is the branch of AI concerned with the design, construction, operation, and use of robots.

- **Applications**: Manufacturing automation, medical robots, exploration (space, underwater), service robots.
- **Challenges**: Motion planning, sensor integration, human-robot interaction.
- **Components**: Actuators, sensors, control systems, AI algorithms.

5. <u>Expert Systems</u>

Expert Systems are AI programs that simulate the decision-making ability of a human expert.

- **Applications**: Medical diagnosis, financial forecasting, troubleshooting systems.
- **Components**: Knowledge base, inference engine, user interface.
- **Advantages**: Consistency, availability, expertise replication.

6. <u>Deep Learning</u>

Neural Networks are computing systems inspired by the biological neural networks of animal brains, and Deep Learning is a subset of ML that uses multi-layered neural networks.

- **Applications**: Image and speech recognition, natural language processing, game playing.
- **Key Concepts**: Backpropagation, convolutional networks, recurrent networks.
- **Technologies**: TensorFlow, PyTorch, Keras.

7. <u>Fuzzy Logic</u>

Fuzzy Logic is a form of many-valued logic that deals with reasoning that is approximate rather than fixed and exact.

- **Applications**: Control systems, decision-making systems, pattern recognition.
- **Advantages**: Handles uncertainty and imprecision well.
- **Examples**: Washing machines, anti-lock braking systems, climate control.

8. Evolutionary Computation

Evolutionary Computation is a family of algorithms for global optimization inspired by biological evolution.

- **Applications**: Optimization problems, artificial life, evolutionary robotics.
- **Techniques**: Genetic algorithms, genetic programming, evolutionary strategies.
- **Key Concepts**: Selection, mutation, crossover, fitness function.

9. <u>Swarm Intelligence</u>

Swarm Intelligence involves the collective behavior of decentralized, self-organized systems, typically made up of a population of simple agents.

- **Applications**: Optimization problems, robotics, network routing.
- **Examples**: Ant colony optimization, particle swarm optimization.

- **Advantages**: Scalability, robustness, flexibility.

10. <u>Cognitive Computing</u>

Cognitive Computing refers to systems that mimic human thought processes in a computerized model.

- **Applications**: Healthcare diagnostics, financial analysis, customer service.
- **Key Features**: Contextual understanding, adaptive learning, interactive capabilities.
- **Technologies**: IBM Watson, cognitive APIs, natural language understanding.